A MINIMAL POINT OF THE SUPPORT IS WEAKLY ASSOCIATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 0-2000 granted to this lane by CONTROL 8.83.

Editorial note. The two references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- A nonzero module has a weakly associated prime. Algebra, Lemma algebra-lemma-weakly-ass-zero, used at proof lines 687-690. Makes WeakAss(F_x) nonempty.
- Weakly associated primes lie in the support. Algebra, Lemma algebra-lemma-weakly-ass-support, used at proof lines 693-695. Places the weakly associated prime inside the support of F_x.

Source reference. Original lemma 0589: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6772–6772. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0588: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6771–6771. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a scheme. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. Let $x \in \mathit{Supp}(\mathcal{F})$ be a point in the support of $\mathcal{F}$ which is not a specialization of another point of $\mathit{Supp}(\mathcal{F})$. Then $x \in \mathit{WeakAss}(\mathcal{F})$. In particular, any generic point of an irreducible component of X is weakly associated to $\mathcal{O}_X$.*

Proof. Since $x \in \mathrm{Supp}(\mathcal{F})$ the module $\mathcal{F}_x$ is not zero. Hence $\mathrm{WeakAss}(\mathcal{F}_x) \subset \mathrm{Spec}(\mathcal{O}_{X,x})$ is nonempty by Algebra, Lemma 0588. On the other hand, by assumption $\mathrm{Supp}(\mathcal{F}_x) = \{\mathfrak{m}_x\}$. Since $\mathrm{WeakAss}(\mathcal{F}_x) \subset \mathrm{Supp}(\mathcal{F}_x)$ (Algebra, Lemma 0589) we see that $\mathfrak{m}_x$ is weakly associated to $\mathcal{F}_x$ and we win. □

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.